

Dimitrios Panagiotis Emmanouilidis

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Software developer working on local LLM inference and the measurement of LLM-driven systems on constrained hardware.

TECHNICAL SKILLS

Languages & OS Python, Linux

Infrastructure & MLOps Docker, Git, VRAM allocation, eGPU configuration, CI/CD concepts

AI & Machine Learning Ollama, LangGraph, local LLM inference, quantised models, RAG, multi-agent orchestration, JSON Schema validation, benchmark design

TECHNICAL PROJECTS

actuation-runner — LLM Action-Payload Failure Benchmark *Sole Author · Public, Aug 2026*

- Open-source benchmark measuring how often LLM-generated action payloads fail before reaching an actuator, across eight failure categories.
- Two 100-trial runs against a local quantised 9B model: 89/100 and 92/100 valid. Every failure across 200 trials was planner token-ceiling truncation.
- 45-test offline suite exercising the real invocation path; caught three defects before the measured run.
- Python, jsonschema, Ollama. Apache 2.0. github.com/dpemmanouilidis/actuation-runner

Offline Local Inference & RAG Application *Lead Developer · In Development*

- Local LLM application for off-grid retrieval, with no cloud API dependency.
- Quantised models tuned for consumer hardware across mobile and eGPU setups.
- RAG pipeline querying offline databases such as Wikipedia.

Two-Agent LLM Pipeline with Deterministic Validation Gate *Dissertation · Early 2026*

- Two-stage LLM pipeline, profiler feeding planner, behind a non-LLM JSON Schema gate, tested in CARLA.
- 4-bit quantised 9B model via Ollama in a 12 GB VRAM budget shared with the simulator.
- Measured end-to-end latency and reported it as prohibitive for real-time control on that hardware.
- Found an idle simulator process halved inference throughput with GPU utilisation unchanged.

Local AI Infrastructure & Orchestration *DevOps & Hardware Ops · Late 2025 – Present*

- Local AI execution server: Beelink GTi13 Ultra with external NVIDIA RTX 5070.
- Dockerised LangGraph and Ollama for reproducible environments.
- Tuned VRAM allocation and hardware bridging.

CERTIFICATIONS

- Python Development Certification (LearnX) — March 2025

EDUCATION

BSc (Hons) Computing (Application Development)

New York College / University of Greater Manchester — Thessaloniki, Greece

Expected July 2026